Exploratory Data Analysis of U.S. Equity Markets

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1 Introduction

This project explores the financial behavior of major publicly traded companies through a comprehensive exploratory data analysis (EDA). The primary goal is to uncover statistical patterns and technical indicators that can inform future predictive modeling using both Machine Learning (ML) and Reinforcement Learning (RL) techniques. By analyzing long-term historical market data, we aim to identify relationships among companies, evaluate their risk and volatility profiles, and assess their suitability for data-driven investment modeling.

The data were sourced from Kaggle's *Stock Market Data (NASDAQ, NYSE, and S&P 500)* collection by Paul Mooney (2023) [2]. This dataset includes daily trading information, `Open`, `High`, `Low`, `Close`, `Adjusted Close`, and `Volume`, for thousands of U.S. equities. It provides a reliable foundation for time-series and cross-sectional analysis.

To ensure coverage of diverse market sectors, eleven representative companies were selected:

- **Technology (NASDAQ)**: Apple (AAPL), Advanced Micro Devices (AMD), Cisco Systems (CSCO), Microsoft (MSFT), Nvidia (NVDA), and Texas Instruments (TXN)
- **Industrial / Logistics (NYSE)**: Emerson Electric (EMR), General Electric (GE),

Honeywell International (HON), International Business Machines (IBM), and United Parcel Service (UPS)

This mix allows the study to contrast high-growth, high-volatility technology firms with the steadier performance of industrial and logistics companies. The resulting analysis highlights how different sectors react to economic events, and it lays the foundation for risk-adjusted model comparison in subsequent stages of the project.

2 Data Preparation and Cleaning

Before conducting exploratory data analysis, the raw dataset required several preprocessing steps to ensure consistency and reliability across all eleven selected companies. Each file was loaded and cleaned using Python's `pandas` library, which provided efficient handling of time-series data. All company datasets were standardized into a unified format, sorted chronologically, and merged into a single comprehensive DataFrame for analysis. Missing values were imputed using a forward-fill method to maintain the temporal integrity of each series, while duplicates and null entries were removed. All numerical variables were converted into floating-point values for accurate computation, and date columns were transformed into `datetime` objects for proper time-based indexing.

Feature Engineering

Once the raw data were cleaned, several essential financial indicators were calculated to describe market behavior more effectively. These derived metrics provide insight into stock performance, volatility, and momentum over time—key inputs for quantitative modeling and machine learning applications.

Daily Return. The daily return measures the percentage change in closing price from one trading day to the next. It quantifies short-term price fluctuations and provides the foundation for most volatility and risk calculations. The daily return, r_t, is expressed as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \tag{1}$$

where P_t represents the closing price on day t, and P_{t-1} is the closing price from the previous trading day. This metric helps identify how much an asset's value changes on a daily basis and whether its price behavior exhibits stability or high sensitivity to market conditions.

Log Return. While the daily return provides a linear rate of change, the log return captures the continuous compounding effect of price movement. Log returns are particularly useful in financial modeling because they are time-additive and more normally distributed, which makes them ideal for regression and machine learning applications. The log return, ℓ_t, is computed as:

$$\ell_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{2}$$

Here, ln denotes the natural logarithm, and the ratio $\frac{P_t}{P_{t-1}}$ reflects the relative change in price. Using log returns smooths extreme percentage fluctuations and allows for better comparison across stocks with different price scales.

Moving Average (MA). To capture short-term and long-term trends, moving averages were calculated for each stock over 7-day and 30-day periods. Moving averages reduce daily volatility by averaging recent prices and help identify momentum shifts over time. The moving average at time t for a window of k days is given by:

$$\mathrm{MA}_k(t) = \frac{1}{k}\sum_{i=0}^{k-1} P_{t-i} \tag{3}$$

This smoothed trend line highlights whether prices are trending upward (bullish) or downward (bearish). Shorter moving averages (e.g., 7-day) respond quickly to recent price changes, while longer ones (e.g., 30-day) provide a broader perspective on overall market direction. Crossovers between the two serve as potential indicators of buying or selling opportunities.

Rolling Volatility. Volatility represents the degree of variation in returns over a specific time window and is one of the most critical indicators of financial risk. To measure it, a 20-day rolling standard deviation was calculated using daily returns, capturing how much returns deviate from their mean within that period. The 20-day volatility, $\sigma_{20}(t)$, is defined as:

$$\sigma_{20}(t) = \sqrt{\frac{1}{20}\sum_{i=1}^{20}(r_{t-i} - \bar{r})^2} \tag{4}$$

where $\bar{r}$ represents the mean of daily returns over the 20-day window. Higher volatility indicates increased uncertainty or price instability, while lower values suggest consistent market performance. Rolling volatility is essential for identifying periods of market stress or calm, which is crucial in model design and risk management.

Relative Strength Index (RSI). The Relative Strength Index (RSI) measures the magnitude of recent price changes to evaluate whether a stock is overbought or oversold. It oscillates between 0 and 100, with values above 70 generally signaling an overbought condition and those below 30 indicating an oversold condition. The RSI over a 14-day period is calculated using the following formulas:

$$RSI_{14} = 100 - \frac{100}{1 + RS} \tag{5}$$

$$RS = \frac{\text{average gain}}{\text{average loss}} \tag{6}$$

RSI helps detect momentum reversals and can indicate when a trend is likely to weaken or reverse. For instance, when RSI consistently hovers near 70, it may suggest that a stock's upward momentum is unsustainable, while values near 30 can signal potential recovery opportunities. This makes RSI a valuable addition for both EDA interpretation and algorithmic trading models.

Final Structured Dataset. After cleaning and feature creation, each company's dataset contained the following columns: `Date`, `Open`, `High`, `Low`, `Close`, `Adjusted Close`, `Volume`, `Return`, `LogReturn`, `MA7`, `MA30`, `Volatility20`, and `RSI14`. All data were aligned by date, ensuring full comparability across companies. The resulting dataset was well-structured and ready for analysis, providing a consistent basis for time-series visualization and cross-company comparison in the EDA phase.

3 Exploratory Data Analysis

3.1 Descriptive Statistics and Return Distributions

Summary statistics for daily returns show that technology stocks, particularly AAPL, AMD, and MSFT, have higher mean returns and larger standard deviations than industrials such as GE and UPS. This pattern reflects the high-growth, high-volatility nature of the tech sector compared to the stable, dividend-driven behavior of industrial companies. Histograms of daily returns approximate a normal distribution centered near zero, with mild positive skew.

Heavy tails among industrials imply rare but impactful market events.

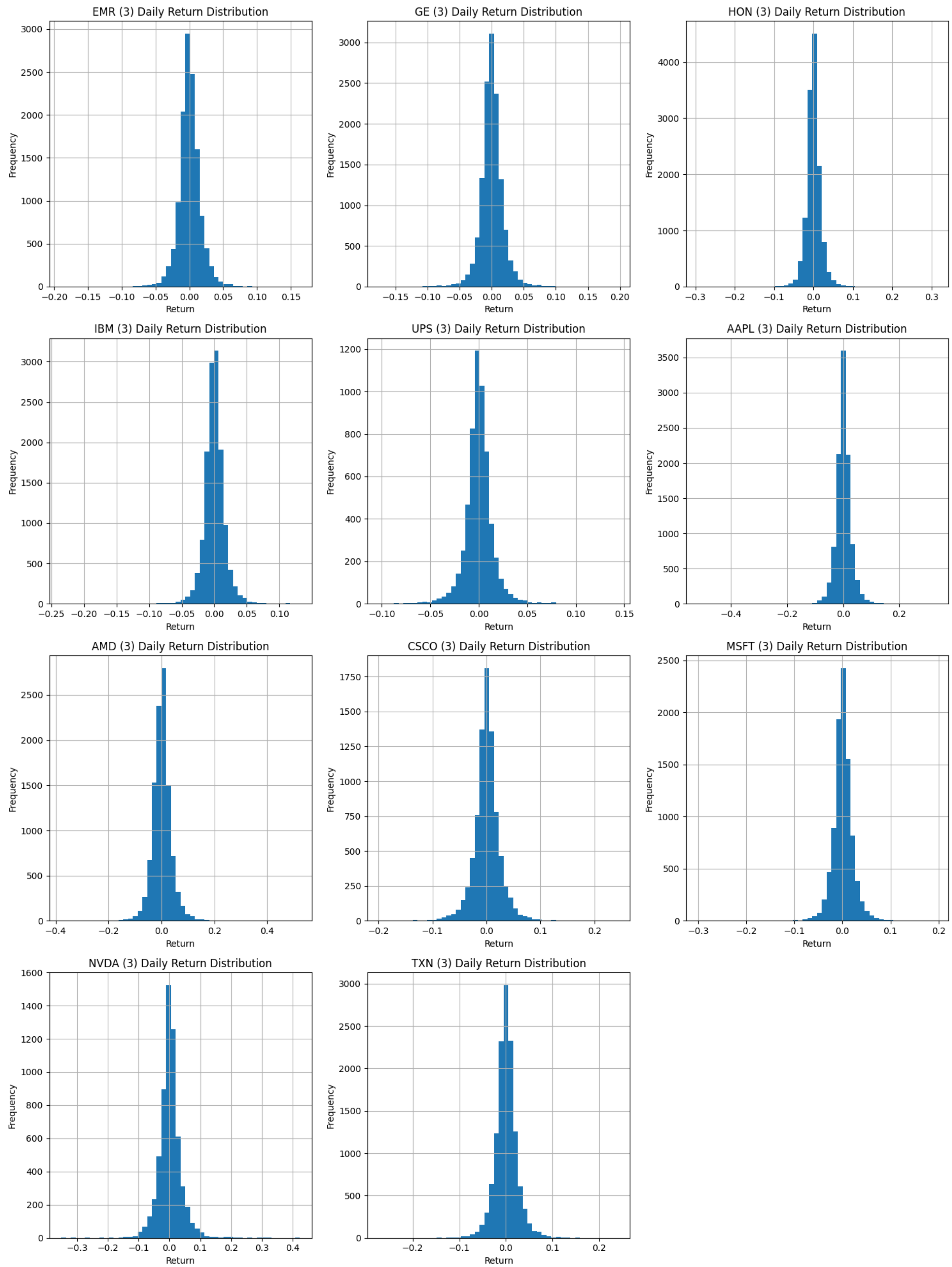

Figure 1: Distribution of daily returns across all selected companies.

Most of the stocks daily return distribution is centered around 0.0 . That means there was a 0 percent or no change in the stock price from day to day. This tells us that for the most part stock prices can be fairly steady day to day.

Next we examined the correlation in price change (i.e daily returns) between the different stocks we analyzed.

3.2 Time-Series Trends and Moving Averages

Figure 2 displays closing prices with 7-day and 30-day moving averages (MA_7 and MA_{30}), which capture both short- and mid-term momentum. Technology firms such as AAPL and MSFT exhibit persistent upward movement punctuated by regular pullbacks, while industrials like GE and IBM demonstrate flatter or cyclical patterns. When the short-term MA_7 crosses above the long-term MA_{30}, it indicates bullish momentum; downward crossovers correspond to bearish shifts. These crossovers act as leading indicators of short-term price movement and are widely used in algorithmic trading systems [1].

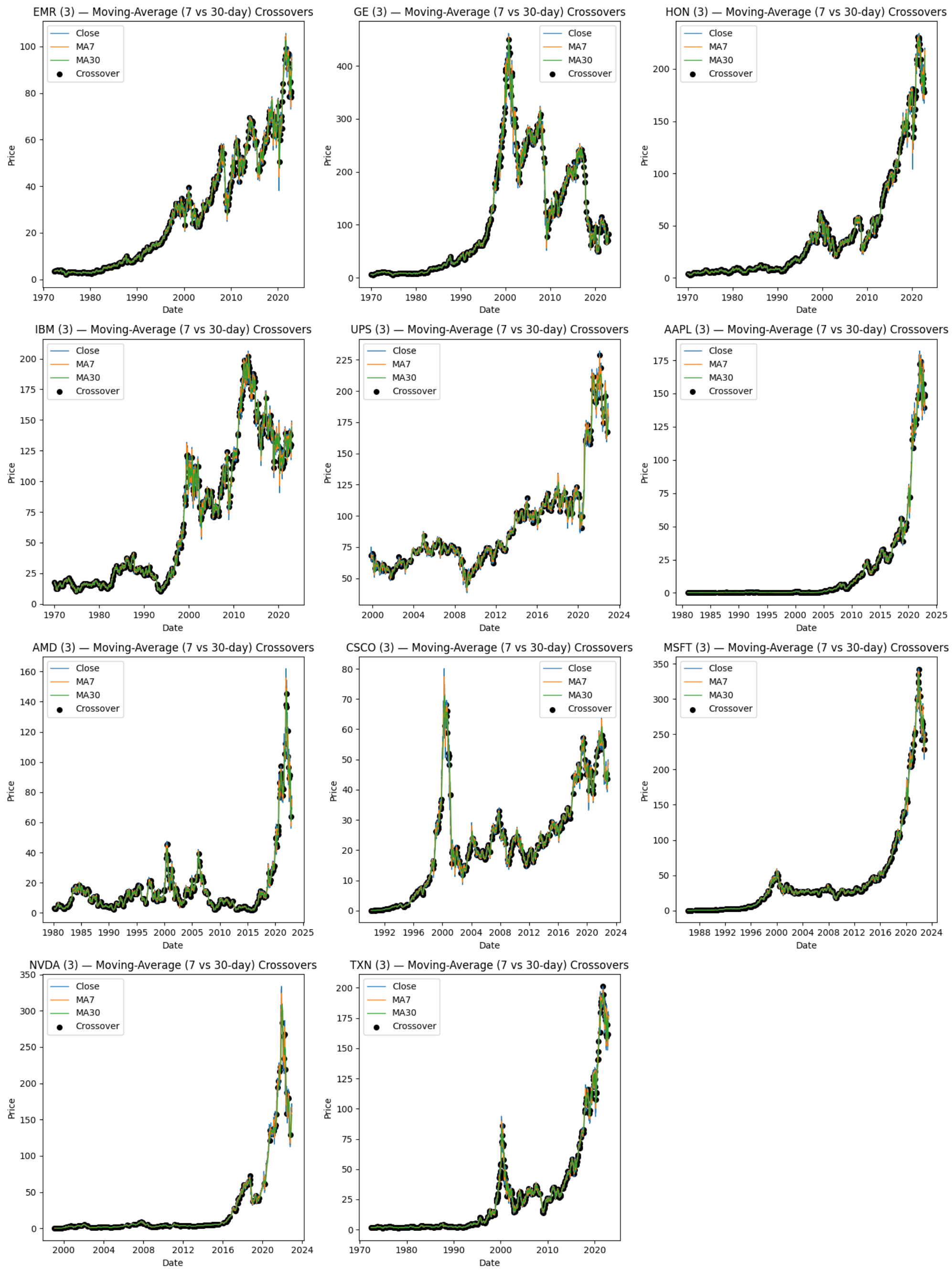

Figure 2: Seven-day and thirty-day moving averages illustrating short- and long-term momentum shifts.

Taking a glimpse at the moving average for 7 and 30 days we see many similarities as well as many crossovers between the two. Looking at AAPL, MSFT, NVDA, HON, they have long, stair-step uptrends where MA7 stays above MA30 for extended stretches; crossovers mainly tag bigger pullbacks (2018, 2022) and resumptions (2020–21, 2023).

When MA7 goes above MA30 in strong-trend names (AAPL/MSFT/NVDA), it tends to precede multi-month advances. When MA7 goes below MA30, it lines up with broader market drawdowns (2000, 2008, 2018 Q4, 2022).

3.3 Relative Strength Index (RSI)

RSI quantifies price momentum on a 0–100 scale (see Figure 3). Values above 70 represent overbought conditions, whereas values below 30 signal oversold conditions. For example, AMD and AAPL frequently dip below 30 during volatile periods and rebound quickly afterward—consistent with short-term mean reversion behavior. Industrial stocks such as UPS and HON maintain RSI within a 40–60 neutral range for extended periods, underscoring their relative stability.

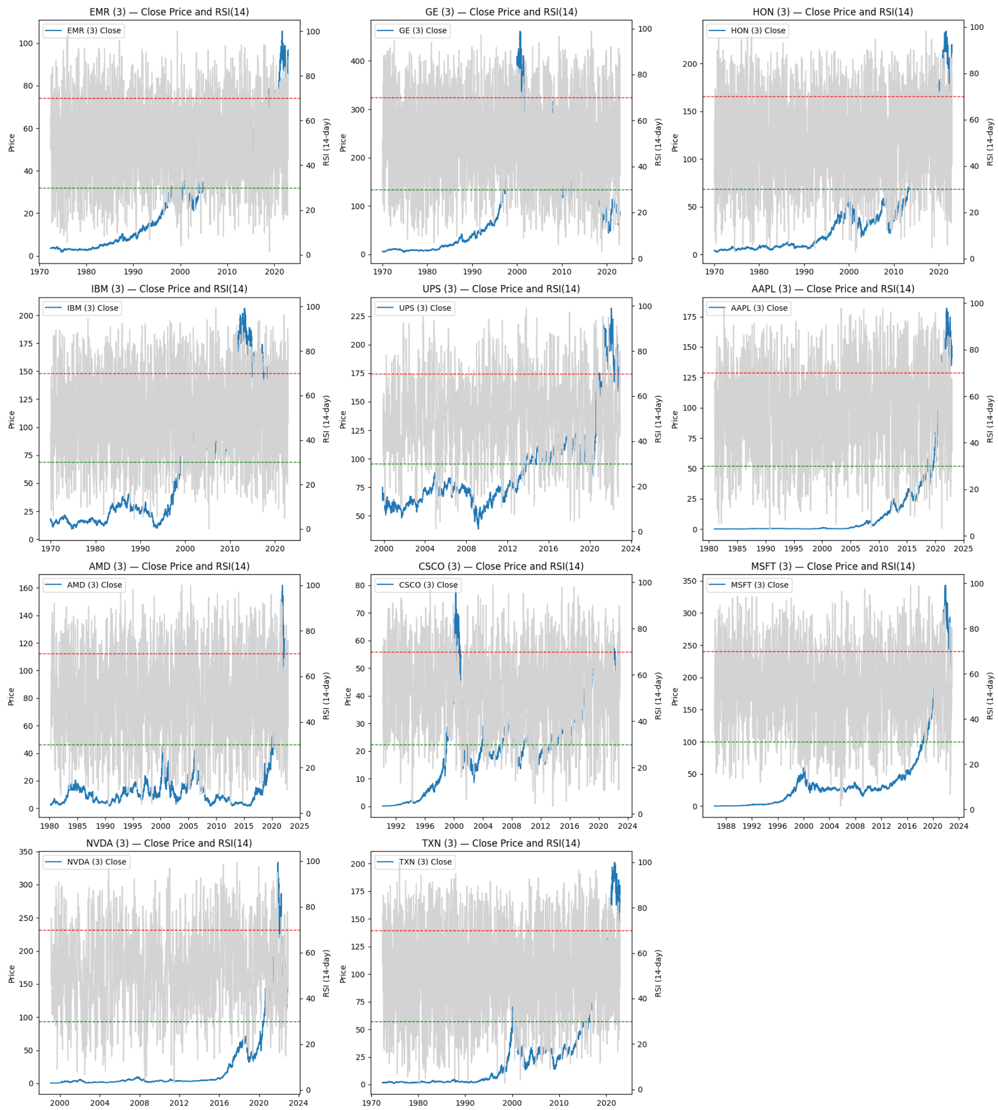

Figure 3: Fourteen-day Relative Strength Index (RSI) across representative companies.

The blue line for all the charts is the closing price over time. The grey line shows the RSI or Relative Strength Index to measure the speed and distance in which the prices have moved.

We will choose a few insights to highlight from these graphs. The first we want to look at is Nvidia (NVDA). Nvidia has had a huge recent bull run so their stock has had a huge climb in recent years. The 14-day RSI valuations for NVDA have repeatedly hit "overbought" territory (above 70) before significant pauses in the uptrend. For example, it recently hit $\approx$78.0 in RSI while the stock was surging. For NVDA, after such extreme RSI, the stock often doesn't collapse immediately, but may pause, consolidate or correct before continuing upward. While this has been the case so far, in different market conditions in the future, if the RSI rises over 70 again, be aware of potential pullback in stock price.

The second stock to take a look at is Texas Instruments (TXN). TXN has had longer, slower growth than the likes of NVDA. The RSI stays mainly in the neutral zone (30-70) except during crashes such as in 2008. If TXN's RSI were to creep above 70, that might serve as a warning that the stock is becoming "stretched" and might be due for a correction or consolidation. Because TXN doesn't habitually hit extreme RSI levels (like NVDA), when it does approach extremes it might be more meaningful — less "normal" than a story stock.

3.4 Volume and Price Relationship

Figure 4 compares closing price with trading volume to identify liquidity-driven trends. Technology companies show significant volume spikes during major product announcements or earnings releases, often aligning with price surges. Industrial stocks maintain steadier trading volume, reflecting consistent institutional participation. High volume concurrent with rising prices implies strong market conviction, while spikes during price declines may indicate profit-taking or investor exit.

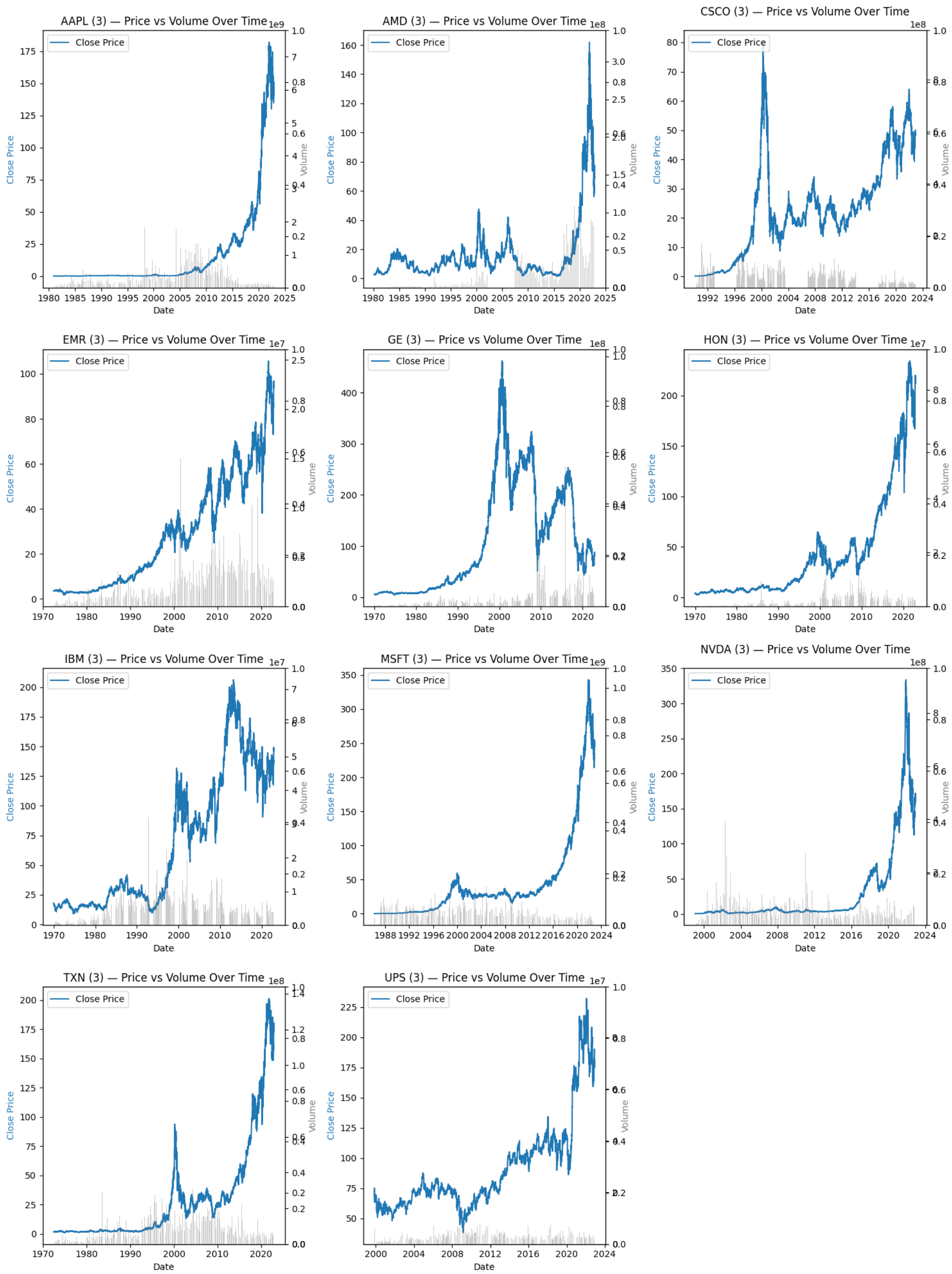

Figure 4: Comparison of closing price and trading volume for selected companies.

These charts look at volume traded throughout the years as well as how the price has changed. We tried to look for points in which high or low volume correlates to price changes. Some of the trends we saw in general were large volume spikes often coinciding with major inflection points—either rapid price runs or sharp corrections. Stocks with steady growth like TXN show moderate volume growth and smoother price curves — indicating more stable investor participation. There's a huge upward trend in price for many of these stocks post 2010.

Looking closely at some of the stocks, CSCO had a high price and consistent volume during the peak of the dot com bubble around 2000. After the bubble crash, both price and volume dropped. Microsoft and UPS have had some of the most consistent trading volume over the span of its time in the stock market.

3.5 Candlestick Visualization

Candlestick charts offer granular insight into intraday market structure by representing open, high, low, and close values for each session (Figure 5). Bullish candles with long lower shadows suggest buying support after early declines, while bearish engulfing formations mark potential trend reversals. Overlaying moving averages (20-, 50-, and 200-day) provides a multi-horizon perspective of momentum and confirms long-term trend direction.

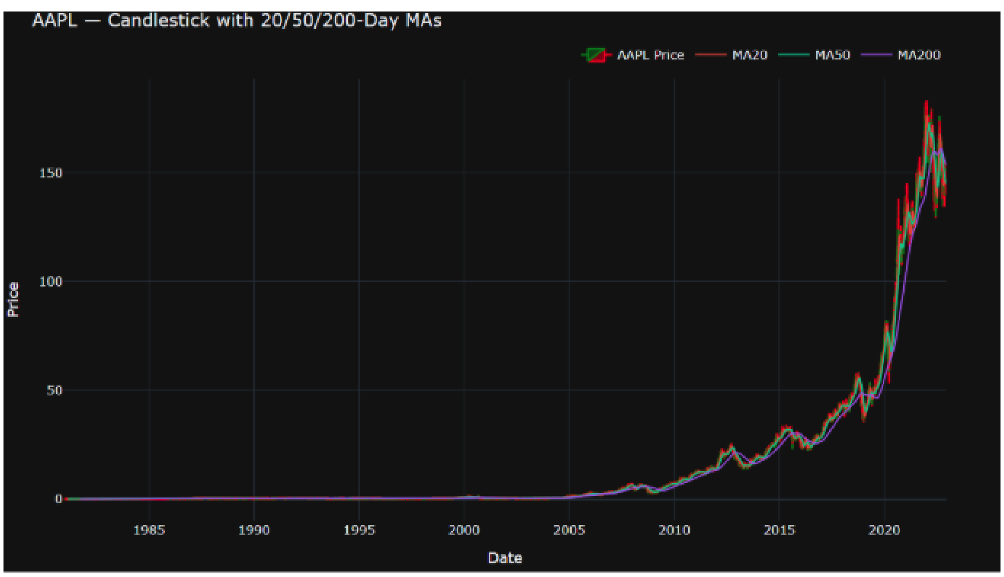

Figure 5: Candlestick chart with moving-average overlays (example: AAPL).

MA20 (fast) $\approx$ 1 month momentum

MA50 (medium) $\approx$ 2–3 months trend check

MA200 (slow) $\approx$ 1 year regime line

Bullish stack: price > MA20 > MA50 > MA200 $\rightarrow$ strongest uptrends

Golden cross (trend): MA50 crosses up through MA200 (medium-term trend turns up)

Momentum cross: MA20 crosses MA50 (earlier but noisier)

Death cross: MA50 below MA200 (downtrend/risk-off)

Moving averages across large time periods can give us insight into the market as a whole even looking at a limited number of stocks. For example, these moving averages show us In broad selloffs (2000, 2008, 2018 Q4, 2022), price falls enough that MA20 then MA50 roll under MA200, producing death crosses nearly everywhere. This is vice versa during recoveries.

3.6 Correlation and Risk–Return Analysis

A Pearson correlation matrix (Figure 6) reveals strong intra-sector relationships among technology firms—particularly AAPL, MSFT, and AMD (r $\approx$ 0.85–0.9). In contrast, industrial firms exhibit moderate correlations (r $\approx$ 0.4–0.5), confirming that diversification across sectors can reduce portfolio volatility. The risk–return scatter plot in Figure 7 further emphasizes this trade-off, where:

$$\text{Sharpe-like Ratio} = \frac{\bar{r}}{\sigma_r}.$$

Technology stocks cluster in the high-return, high-volatility quadrant, while industrials occupy the low-risk, low-return zone, aligning with expected sector characteristics [3].

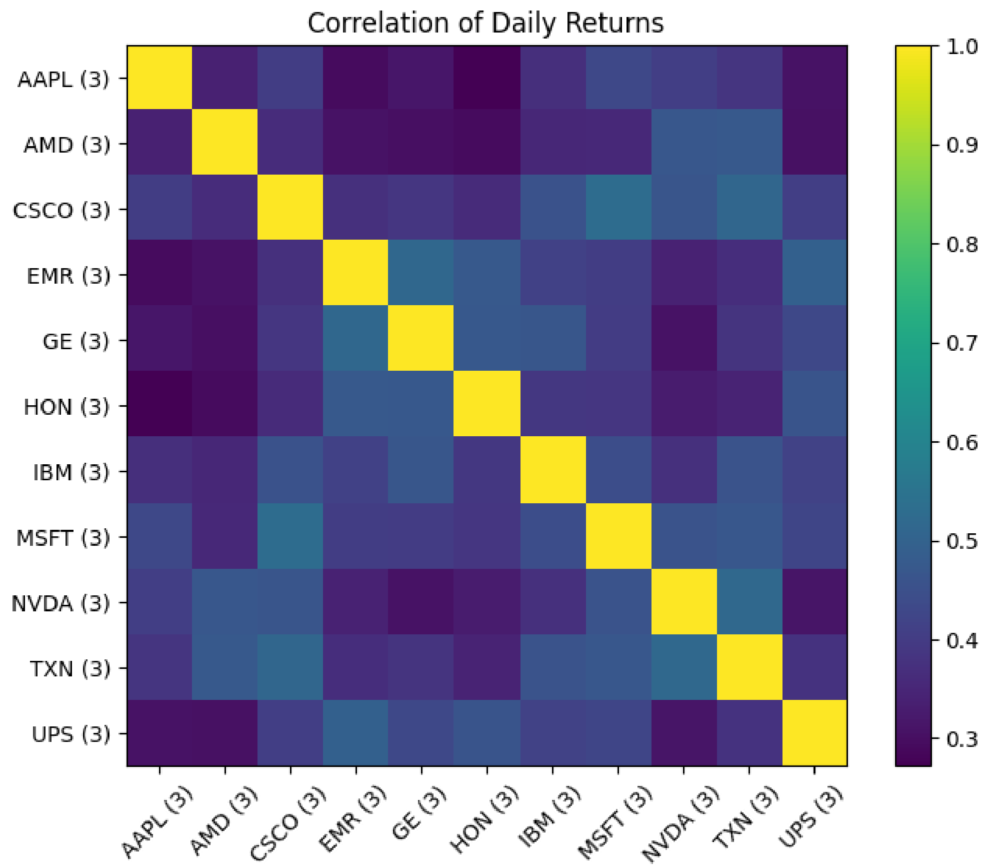

Figure 6: Correlation Matrix of Daily Returns

We found that changes in the daily price have basically weak to moderate correlation (but mostly weak) between the stocks we chose to analyze. There was no general trend in price changes between stocks at the short term daily level.

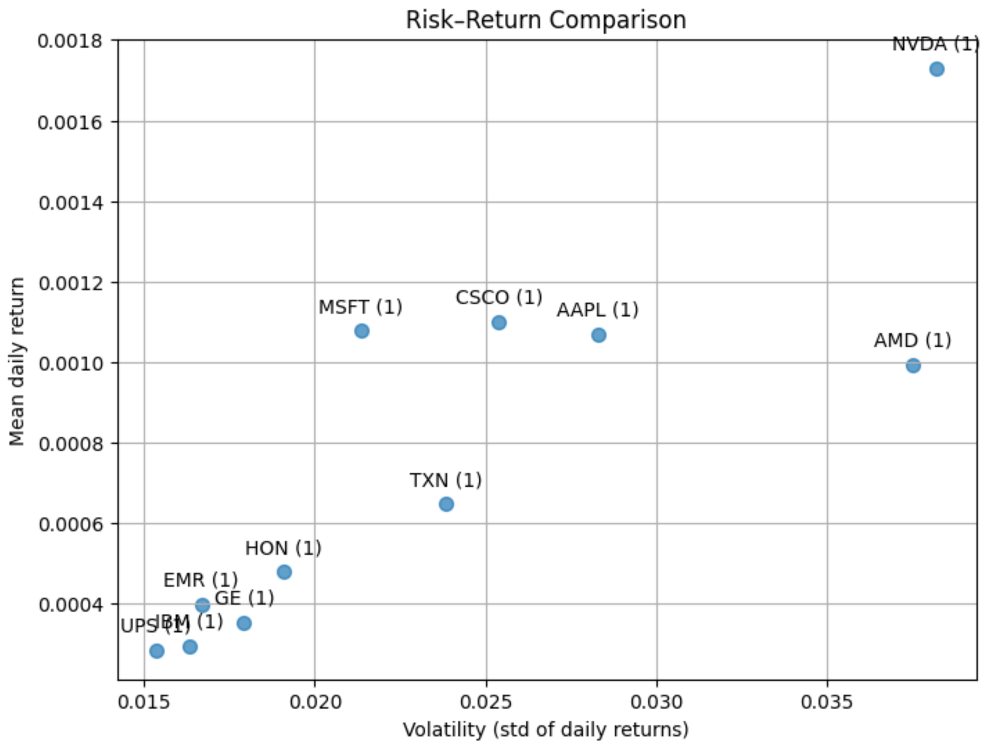

Figure 7: Risk Return Comparison looking at Volatility vs. Daily Returns

This figure looks at how risky a stock is versus how profitable of an investment it is. NVDA has the highest risk, but the highest reward. EMR, UPS, IBM, GE, HON are all relatively lower risk but lower average daily returns. AAPL, CSCO, MSFT have moderate returns and moderate risk. AMD has high volatility, but only medium returns. Finally, TXN has moderate volatility but mid- to low-performance returns.

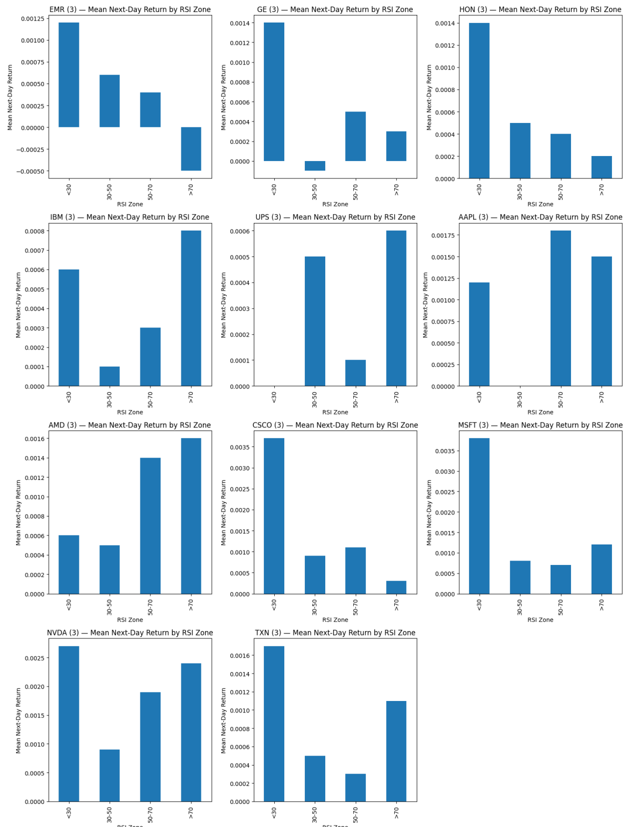

Figure 8: Next Day Return by RSI Bin

We also took a look at how the RSI zone for that day affected the return on average. A positive bar means that, on average, the stock's price rose the next day when RSI was in that zone. A negative bar means it tends to fall the next day.

Some stocks reverted to the mean at RSI less than 30 meaning that the day after their RSI is less than 30, the stock rebounded. These stocks included EMR and HON. Momentum stocks like APPL and NVDA have positive returns even when their RSI is greater than 70.

4 Conclusion

The EDA demonstrates clear distinctions between growth-oriented technology companies and stable industrial corporations. Technology stocks exhibit higher volatility, stronger short-term momentum, and tighter intra-sector correlation. Industrial firms provide lower volatility and smoother RSI and volume profiles, contributing diversification benefits.

Overall, the dataset is complete, well-structured, and ready for subsequent modeling stages. These EDA results validate the data integrity and reveal technical indicators—such as moving averages, RSI, and volatility—that will serve as essential predictive features in the project's machine learning and reinforcement learning models.

References

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[2] Paul Mooney. Stock market data (nasdaq, nyse, s&p 500). https://www.kaggle.com/datasets/paultimothymooney/stock-market-data, 2023.

[3] William F. Sharpe. Mutual fund performance. *Journal of Business*, 39(1):119–138, 1966.